

Graphs - Set 3 Eulerian and Hamiltonian Graphs

1. Find a Hamiltonian cycle in the graph formed by the vertices and edges of a cube (it is known that every platonic solid, considered as a graph, is hamiltonian)
2. Next year, tourists A and B (see Seminar 9) intend to visit another place, where interesting places and the roads joining them are representing by the graph whose adjacency list is

0	1	2	3	4	5	6	7	8
1	0	1	0	3	0	1	0	1
3	2	3	2	5	4	5	2	3
5	6	7	4		6	7	6	5
7	8		8		8		8	7

It is possible to find routes for them which satisfy the requirements in the analogous problem in Seminar 9?

3. A mouse intends to eat a $3 \times 3 \times 3$ cube of cheese. Being tidy-minded, it begins at a corner and eats the whole of $1 \times 1 \times 1$ cube before going on to an adjacent one. Can the mouse end in the center?
4. Consider the map in Figure 1 of 7 towns and roads connecting them.

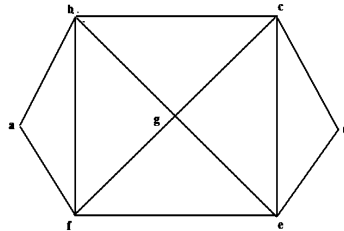


Figure 1: Problem 4.

A highway engineer (E) wishes to inspect all the roads, whilst an inspector (H) wishes to dine in a restaurant in each town. Each wishes to achieve their objective as efficient as possible. So, E states "I wish, if possible, to visit every road once and only once and return to my starting point", whilst H says "I wish, if possible, to visit each town once and only once and to return to my starting point". Find possible solutions for both of them.

5. Which of the graphs in Figure 2 are Eulerian and which are Hamiltonian? In case they are Hamiltonian, give an example of Hamiltonian cycle and in case they are Eulerian give an example of an Eulerian tour.
6. Draw the specified graph or prove that it does not exist:
 - a. An 8-vertex simple graph with more than 8 edges that is both Hamiltonian and Eulerian.
 - b. An 8-vertex simple graph with more than 8 edges that is Eulerian but not Hamiltonian.

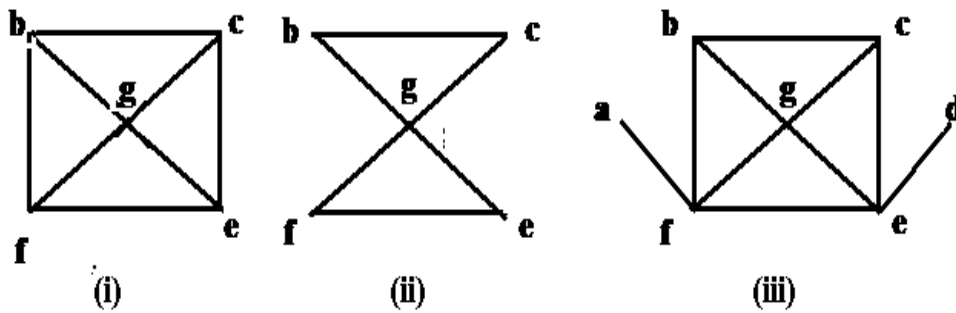


Figure 2: Problem 5.

- c. An 8-vertex simple graph with more than 8 edges that is Hamiltonian but not Eulerian.
- d. An 8-vertex simple Hamiltonian graph that does not satisfy the conditions of Ore's theorem.
- e. A 6-vertex simple graph with 10 edges that is not Hamiltonian.
- 7. For the graphs in Figure 3, apply Fleury's algorithm to construct an Euler tour.

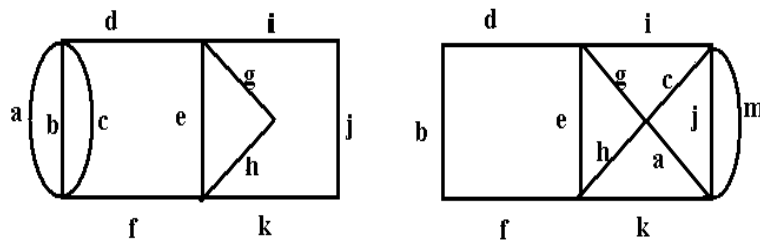


Figure 3: Problem 7.

- 8. Apply Fleury's algorithm constructing an open Eulerian trail in the graphs in Figure 4.

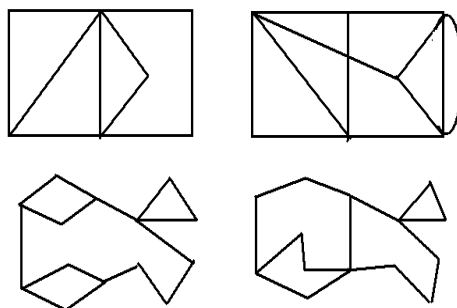


Figure 4: Problem 8.

- 9. The following rules are based on the simple observation that any Hamiltonian cycle must contain exactly two edges incident on each vertex:

Rule 1: If a vertex v has degree 2, then both of its incident edges must be part of any hamiltonian cycle.

Rule 2: During the construction of a hamiltonian cycle, no cycle can be founded until all the vertices have been visited.

Rule 3: If during the construction of a hamiltonian cycle two of the edges incident on a vertex v are shown to be required, then all other incident edges can be deleted. The strategy for applying these rules is to begin a construction of a Hamiltonian cycle and show at some point during the construction that it is impossible to proceed any further.

Prove in the way described above that the graphs in Figure 5 are not Hamiltonian.

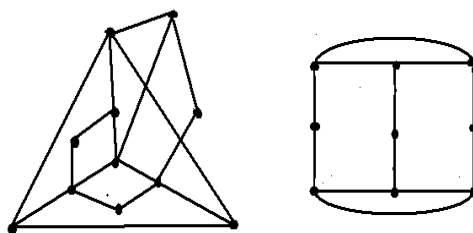


Figure 5: Problem 9.

Answers, hints: **1.** $ABCC'B'A'D'DA$ **3** no **5.**(i) H, not E (ii) E not H (iii) not E, not H.